

# Unit 1: Function Analysis

Precalculus

Student

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## 1.1 Functions and Function Notation

### Objectives

- Determine whether a relation represents a function.
- Use function notation to evaluate functions.
- Represent functions using tables, mapping diagrams, and graphs.

#### Relations and Functions

A **relation** is \_\_\_\_\_. The set of all first coordinates is the **domain**, and the set of all second coordinates is the **range**.

A **function** is a relation that assigns to each element in the domain \_\_\_\_\_ in the range.

**Key Idea:** One input  $\rightarrow$  one output. Two different inputs *can* share the same output, but one input *cannot* produce two different outputs.

#### Example 1

Determine whether each relation represents a function.

a)  $\{(1, 5), (2, 6), (3, 7), (4, 8)\}$

b)  $\{(1, 5), (1, 6), (3, 7), (4, 8)\}$

#### Mapping Diagrams

A **mapping diagram** uses arrows to show how inputs in the domain relate to outputs in the range.

- **Function:** Each input in the domain has \_\_\_\_\_ arrow leaving it.
- **Not a Function:** If any input in the domain has \_\_\_\_\_ arrows leaving it.

#### Function Notation

Instead of writing  $y$ , we write \_\_\_\_\_. Read as “\_\_\_\_\_” or “the value of  $f$  at  $x$ .”

The letter  $f$  names the function. The letter  $x$  is the input.

#### Example 2

Evaluate  $f(x) = x^2 - 3x + 5$ . Find each value.

a)  $f(2)$

b)  $f(-1)$

c)  $f(a)$

d)  $f(a + h)$

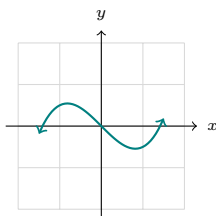
### Vertical Line Test (VLT)

A set of points in a coordinate plane is the graph of a function if and only if every vertical line intersects the graph in \_\_\_\_\_.

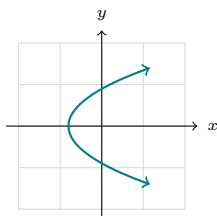
#### Example 3

Determine whether each graph represents a function using the vertical line test.

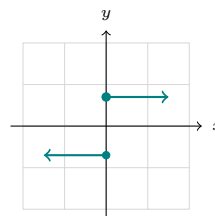
a) Wavy curve



b) Sideways parabola



c) Piecewise step



#### Example 4

Use the table of values for  $f(x)$  to evaluate the expressions.

| $x$    | 1 | 2 | 3 | 4 | 5 |
|--------|---|---|---|---|---|
| $f(x)$ | 8 | 6 | 7 | 6 | 8 |

a) Evaluate  $f(1)$

b) Evaluate  $f(3)$

c) Solve  $f(x) = 6$

#### Example 5

The height of a baseball hit is modeled by the function  $f(x) = -0.0032x^2 + x + 3$ , where  $x$  is the horizontal distance from home plate in feet. Will the baseball clear a 10-foot fence located 300 feet from home plate?

### One-to-One Functions

A function is **one-to-one** if each element of the range corresponds to \_\_\_\_\_ of the domain.

Equivalently, if  $f(a) = f(b)$ , then \_\_\_\_\_. Different inputs must produce different outputs.

**Example 6**

Determine which of the following equations define  $y$  as a function of  $x$ .

a)  $5x - 3y^2 = 9$

b)  $3y^3 + x = 1$

c)  $y = 5x^2 - 6x$

d)  $x^2 = |y| + 6x$

*Homework:* Section 1.1 Homework

**1.2 Domain and Range****Objectives**

- Find the domain and range of functions algebraically and from graphs.
- Write intervals in interval and set-builder notation.
- Evaluate and graph piecewise-defined functions.

**Domain and Range**

The **domain** of a function is the set of all possible \_\_\_\_\_ (typically  $x$ ) for which the function is defined and produces a real number output.

The **range** of a function is the set of all possible \_\_\_\_\_ (typically  $y$ ) that the function can produce.

**Representing Intervals**

We can write sets of numbers using inequalities, set-builder notation, or interval notation.

- **Parentheses**  $()$ : Used when an endpoint is \_\_\_\_\_ (strictly less than, strictly greater than, or at  $\pm\infty$ ).
- **Brackets**  $[]$ : Used when an endpoint is \_\_\_\_\_ (less than or equal to, or greater than or equal to).

| Description                         | Inequality             | Set-Builder                   | Interval            |
|-------------------------------------|------------------------|-------------------------------|---------------------|
| All real numbers                    | $-\infty < x < \infty$ | $\{x \mid x \in \mathbb{R}\}$ | $(-\infty, \infty)$ |
| Greater than or equal to $a$        | $x \geq a$             | $\{x \mid x \geq a\}$         | $[a, \infty)$       |
| Strictly between $a$ and $b$        | $a < x < b$            | $\{x \mid a < x < b\}$        | $(a, b)$            |
| Closed interval between $a$ and $b$ | $a \leq x \leq b$      | $\{x \mid a \leq x \leq b\}$  | $[a, b]$            |

**Algebraic Domain Restrictions**

Unless specified otherwise, we assume the domain is all real numbers. We must restrict the domain algebraically in two common cases:

1. **Division by Zero:** The denominator of a fraction cannot equal zero. If  $f(x) = \frac{g(x)}{h(x)}$ , we set \_\_\_\_\_.
2. **Even Roots of Negatives:** The radicand (expression inside) of an even-index root must be non-negative. If  $f(x) = \sqrt[2k]{g(x)}$ , we set \_\_\_\_\_.

*Note:* Polynomials and odd roots have no algebraic restrictions, so their domain is  $(-\infty, \infty)$ .

### Example 1

Find the domain of each function in interval notation.

a)  $f(x) = \frac{x+1}{2-x}$

b)  $g(x) = \sqrt{7-x}$

c)  $h(x) = \frac{5}{\sqrt{x-3}}$

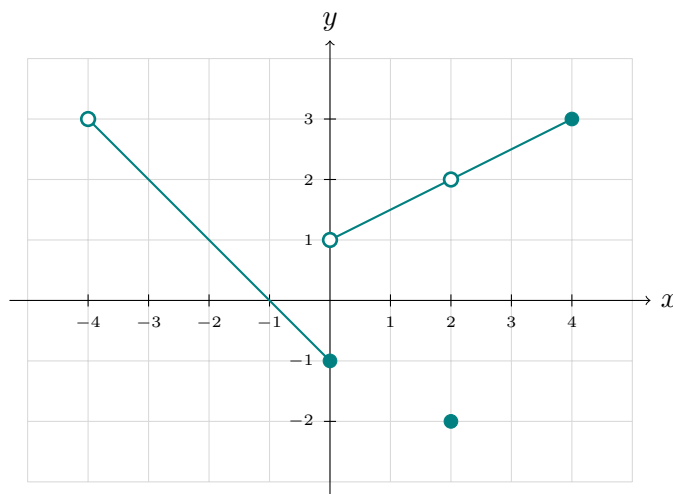
### Domain and Range from Graphs

To determine domain and range from a graph:

- **Domain:** Scan the graph horizontally (left to right) along the  $x$ -axis.
- **Range:** Scan the graph vertically (bottom to top) along the  $y$ -axis. Check local peaks and valleys!
- **Endpoints:** A **closed circle** • indicates the value is \_\_\_\_\_ ( $[]$ ). An **open circle** ○ indicates the value is \_\_\_\_\_ ( $()$ ).

### Example 2

State the domain and range of the function shown in the graph below in interval notation.



### Piecewise Functions

A **piecewise function** is defined by more than one formula, where each formula applies to a different part of the domain.

To evaluate  $f(a)$ :

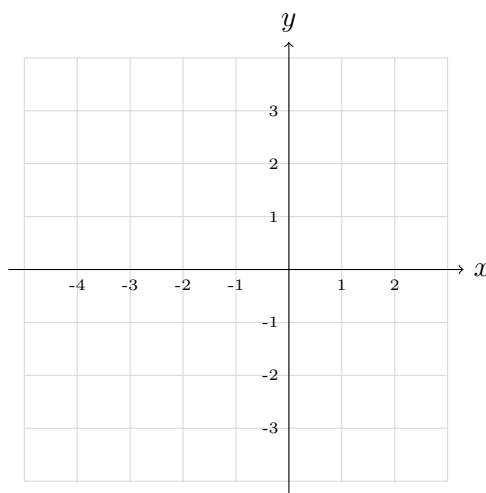
1. Check which interval  $a$  falls into.
2. Plug  $a$  into **only** the equation for that interval.

### Example 3

Consider the piecewise function:

$$f(x) = \begin{cases} x + 1, & \text{if } x < -2 \\ -2x - 3, & \text{if } x \geq -2 \end{cases}$$

- a) Evaluate  $f(-3)$ .                      b) Evaluate  $f(-2)$ .                      c) Graph the function and state its range.



### Example 4

A cellular provider charges \$30 per month for a basic plan, plus \$10 per gigabyte (GB) of data used, capped at a maximum of 5 GB.

- Write the monthly cost  $C(g)$  as a function of data usage  $g$ .
- State the domain and range of this function in this real-world context.

*Homework:* Section 1.2 Homework

## 1.3 Rates of Change and Behavior of Graphs

### Objectives

- Calculate the average rate of change of a function.

- Identify intervals where a function is increasing, decreasing, or constant.
- Locate local and absolute extrema on a graph.

### Average Rate of Change

The **average rate of change** between two input values is the change in the output values divided by the change in the input values.

$$\text{Average Rate of Change} = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

- **Geometric Interpretation:** The average rate of change represents the \_\_\_\_\_ passing through the points  $(x_1, f(x_1))$  and  $(x_2, f(x_2))$  on the graph of the function.
- **Units:** The units of the average rate of change are always \_\_\_\_\_ (e.g., miles per hour, dollars per year).

### Example 1

The table shows the average price of a gallon of gasoline in the US between 2005 and 2020. Find the average rate of change of gas prices between the given years.

| Year ( $t$ )     | 2005   | 2008   | 2012   | 2016   | 2020   |
|------------------|--------|--------|--------|--------|--------|
| Price ( $P(t)$ ) | \$2.30 | \$3.27 | \$3.64 | \$2.14 | \$2.60 |

a) Between 2005 and 2012

b) Between 2012 and 2020

### Example 2

Find the average rate of change of  $f(x) = x^2 - \frac{1}{x}$  on the interval  $[2, 4]$ .

### Increasing, Decreasing, and Constant Behavior

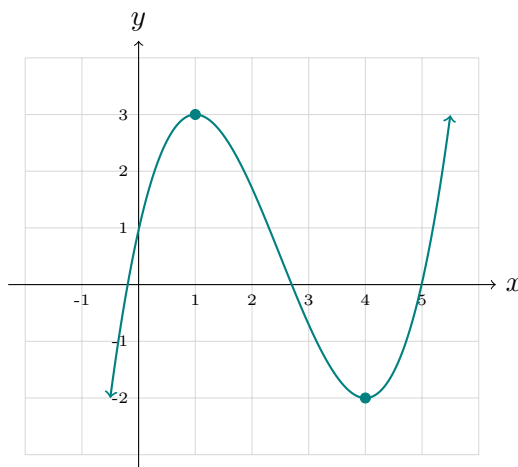
Read the graph from **left to right**:

- **Increasing:** Graph rises from left to right. Outputs increase as inputs increase:  $f(x_1) < f(x_2)$  when  $x_1 < x_2$ .
- **Decreasing:** Graph falls from left to right. Outputs decrease as inputs increase:  $f(x_1) > f(x_2)$  when  $x_1 < x_2$ .
- **Constant:** Graph is flat. Outputs do not change:  $f(x_1) = f(x_2)$  for all  $x$ .

**Critical Rule:** When reporting intervals where a function is increasing, decreasing, or constant, always write the intervals in terms of \_\_\_\_\_ using open intervals (parentheses).

### Example 3

Identify the intervals where the function graphed below is increasing and/or decreasing.



### Local and Absolute Extrema

- A **local (relative) maximum** is a point higher than any nearby points (a hill peak).
- A **local (relative) minimum** is a point lower than any nearby points (a valley bottom).
- An **absolute (global) maximum** is the highest output value over the function's entire domain.
- An **absolute (global) minimum** is the lowest output value over the function's entire domain.

**Domain Rule:** On a closed interval  $[a, b]$ , absolute extrema must exist and can occur at \_\_\_\_\_. However, local extrema can only occur on the interior of the domain, never at endpoints.

### Example 4

Using the graph from Example 3, identify all local and absolute extrema on the domain  $(-\infty, \infty)$ .

**Example 5**

Suppose the domain of the function from Example 3 is restricted to the closed interval  $[0, 5]$ , where  $f(0) = 1$  and  $f(5) = 1$  (using the endpoints). Find the local and absolute extrema.

*Homework:* Section 1.3 Homework

**1.4 Composition of Functions****Objectives**

- Combine functions algebraically and determine their domains.
- Evaluate composite functions numerically and algebraically.
- Find the domain of a composite function.
- Decompose a composite function into component functions.

**Arithmetic Operations on Functions**

We can combine functions using standard arithmetic operations. Let  $f$  and  $g$  be functions:

- **Sum:**  $(f + g)(x) = f(x) + g(x)$
- **Difference:**  $(f - g)(x) = f(x) - g(x)$
- **Product:**  $(f \cdot g)(x) = f(x) \cdot g(x)$
- **Quotient:**  $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$

**Overlapping Domains:** The domain of  $f \pm g$ ,  $f \cdot g$ , and  $\frac{f}{g}$  is the \_\_\_\_\_ (overlap) of the domain of  $f$  and the domain of  $g$ . For the quotient function, we must also exclude any values of  $x$  where \_\_\_\_\_.

**Example 1**

Given  $f(x) = 3x^2 + 7x$  and  $g(x) = 2x^2 - x - 1$ , find  $(f - g)(x)$ .

**Example 2**

Given  $f(x) = 3x^2 - 2x + 1$  and  $g(x) = x - 4$ , find  $\left(\frac{f}{g}\right)(x)$  and state its domain in interval notation.

**Composition of Functions**

When we compose functions, we nest one function inside another:

- **$f$  composed with  $g$ :**  $(f \circ g)(x) = f(g(x))$  (read as “ $f$  of  $g$  of  $x$ ”)
- **$g$  composed with  $f$ :**  $(g \circ f)(x) = g(f(x))$  (read as “ $g$  of  $f$  of  $x$ ”)

**Order Matters:** Composition is not commutative: in general,  $(f \circ g)(x) \neq (g \circ f)(x)$ .

**Strategy:** Always evaluate a composition from the \_\_\_\_\_.

**Example 3**

Given  $f(x) = 2x + 3$  and  $g(x) = x^2 - 1$ , evaluate the following expressions:

a)  $f(g(1))$

b)  $g(f(2))$

**Example 4**

Given  $f(x) = x - 3$  and  $g(x) = 2x^2 - 1$ , find the following algebraic composite expressions:

a)  $(f \circ g)(x)$

b)  $(g \circ f)(x)$

**Domain of a Composite Function**

To find the domain of the composite function  $f(g(x))$ :

1. Find the domain of the **inner function**  $g(x)$ . Exclude any restricted values.
2. Form the **composite expression**  $f(g(x))$ , simplify, and identify any additional restrictions on this final expression.
3. Combine the exclusions. The final domain is where both conditions are satisfied.

**Example 5**

Find the domain of  $(f \circ g)(x)$  given  $f(x) = \frac{1}{x+2}$  and  $g(x) = \frac{4}{x-1}$ .

**Decomposition of Functions**

To **decompose** a complex function  $h(x)$  means to find an outer function  $f(x)$  and an inner function  $g(x)$  such that  $h(x) = f(g(x))$ .

Look for group expressions inside parentheses, square roots, denominators, or exponents.

**Example 6**

Find functions  $f(x)$  and  $g(x)$  such that  $h(x) = f(g(x))$  for each function.

a)  $h(x) = \sqrt{5x-4}$

b)  $h(x) = \frac{4}{(x+2)^3}$

*Homework:* Section 1.4 Homework

**1.5 Transformation of Functions****Objectives**

- Identify translations, reflections, and dilations from their equations.
- Apply the correct order of transformations to graph functions.
- Classify functions as even, odd, or neither algebraically and graphically.

**Parent Functions**

You must be familiar with the following basic parent functions:

- |                                  |                                       |                                          |
|----------------------------------|---------------------------------------|------------------------------------------|
| • <b>Linear:</b> $f(x) = x$      | • <b>Cubic:</b> $f(x) = x^3$          | • <b>Square Root:</b> $f(x) = \sqrt{x}$  |
| • <b>Quadratic:</b> $f(x) = x^2$ | • <b>Absolute Value:</b> $f(x) =  x $ | • <b>Cube Root:</b> $f(x) = \sqrt[3]{x}$ |

**Three Flavors of Transformation**

Let  $f(x)$  be a parent function. We can transform it in three ways:

1. **Translation (Shift):** Slides the graph horizontally by  $h$  or vertically by  $k$ .

$$g(x) = f(x - h) + k$$

*Note:*  $f(x - h)$  shifts \_\_\_\_\_ (opposite of sign) and  $f(x) + k$  shifts \_\_\_\_\_

(behaves as expected).

2. **Reflection:** Flips the graph across the axes.

- $-f(x)$ : Negates  $y$ -values  $\rightarrow$  reflects over the \_\_\_\_\_.
- $f(-x)$ : Negates  $x$ -values  $\rightarrow$  reflects over the \_\_\_\_\_.

3. **Dilation (Stretch/Compression):** Changes the size/shape of the graph.

- $c \cdot f(x)$  (vertical): Stretch if  $c > 1$ , compress if  $0 < c < 1$ .
- $f(kx)$  (horizontal): Compress if  $k > 1$ , stretch if  $0 < k < 1$  by a factor of  $\frac{1}{k}$ .

**Core Principle:** Changes **outside** affect  $y$  (vertical, normal behavior). Changes **inside** affect  $x$  (horizontal, opposite behavior).

### Example 1

Describe the transformations from the parent function for each of the following:

a)  $g(x) = |x - 1| + 12$

b)  $h(x) = -\sqrt{x + 8}$

c)  $f(x) = 3(x - 7)^3$

d)  $h(x) = \sqrt[3]{2x - 9}$

### Order of Transformations: D.R.S.

When multiple transformations are applied, follow the **D.R.S.** order:

$$\mathbf{D} \rightarrow \mathbf{R} \rightarrow \mathbf{S} \quad (\text{Dilate} \rightarrow \text{Reflect} \rightarrow \text{Shift})$$

Applying transformations in this order ensures that shifts are performed on the dilated and reflected graph correctly.

### Even and Odd Functions

- **Even Function:** Symmetric with respect to the \_\_\_\_\_. Algebraic test: \_\_\_\_\_.
- **Odd Function:** Symmetric with respect to the \_\_\_\_\_. Algebraic test: \_\_\_\_\_.

### Example 2

Classify each function as even, odd, or neither algebraically.

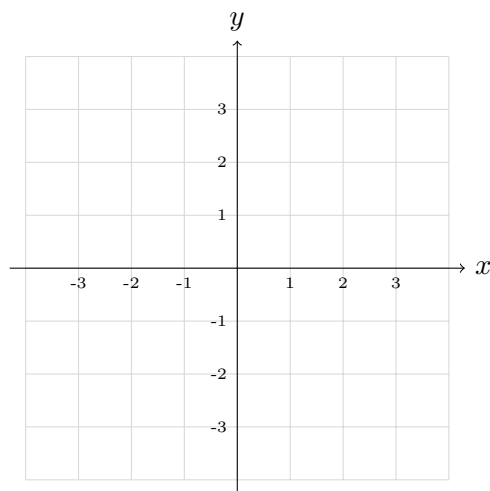
a)  $f(x) = x^3 - x$

b)  $g(x) = x^2 + 2$

c)  $h(x) = x^3 - 1$

**Example 3**

Describe the transformations, map the points, and graph  $y = -2|x| + 1$ .



**Transformations (D.R.S.):**

- D: \_\_\_\_\_
- R: \_\_\_\_\_
- S: \_\_\_\_\_

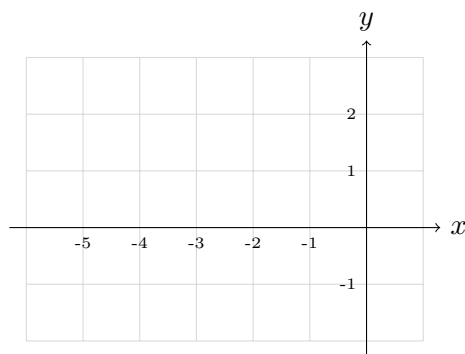
**Mapping**

**Table:**

| $x$ | Parent $ x $ | Transformed |
|-----|--------------|-------------|
| -2  | 2            | _____       |
| -1  | 1            | _____       |
| 0   | 0            | _____       |
| 1   | 1            | _____       |
| 2   | 2            | _____       |

**Example 4**

Describe the transformations, map the points, and graph  $y = \sqrt{-2(x+3)} - 1$ .



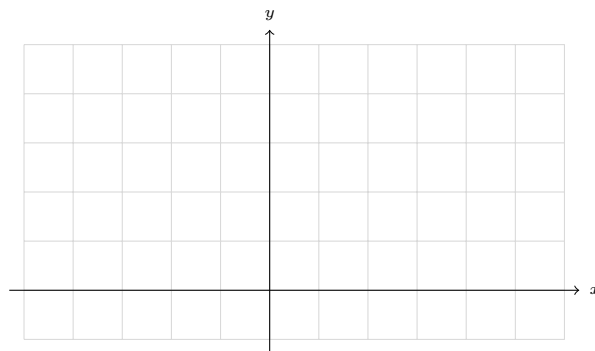
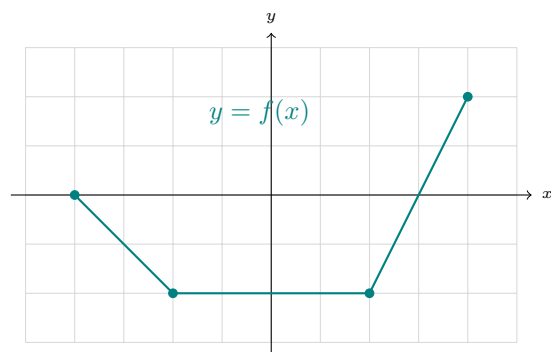
**Transformations (D.R.S.):**

- D: \_\_\_\_\_
- R: \_\_\_\_\_
- S: \_\_\_\_\_

| Mapping Table: | Parent $(x, \sqrt{x})$ | Transformed |
|----------------|------------------------|-------------|
|                | (0, 0)                 | _____       |
|                | (1, 1)                 | _____       |
|                | (4, 2)                 | _____       |

**Example 5**

Given the graph of  $y = f(x)$  below, graph  $g(x) = -\frac{1}{2}f(x-1) + 3$ .



*Homework:* Section 1.5 Homework

**1.6 Absolute Value Functions****Objectives**

- Graph absolute value functions using vertex and transformations.
- Solve absolute value equations algebraically.

- Solve absolute value inequalities and graph their solutions.

### The Absolute Value Function

The parent function  $f(x) = |x|$  represents the distance of  $x$  from the origin. It is defined piecewise as:

$$f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

- **Domain:**  $(-\infty, \infty)$
  - **Vertex:** The corner point where the graph changes direction, which is at  $(0, 0)$ .
- Range:**  $[0, \infty)$

### Note

Because squaring removes a sign and the principal square root is never negative,  $\sqrt{x^2} = |x|$  for every real number  $x$ . This identity lets you rewrite an even-power-under-a-root expression as an absolute value.

### Graphing with Transformations

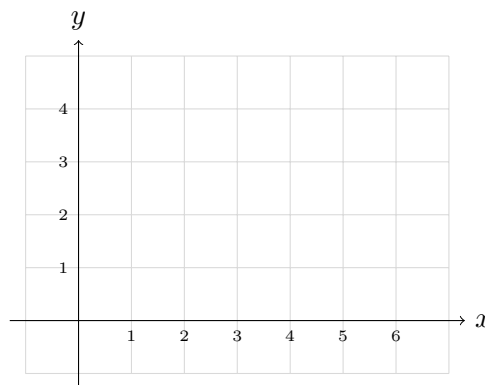
The general form of a transformed absolute value function is:

$$f(x) = a|x - h| + k$$

- **Vertex:** Shifted to \_\_\_\_\_.
- **Orientation:** Opens up if  $a > 0$ ; opens down if  $a < 0$ .
- **Branch Slopes:** The right branch has slope  $a$ , and the left branch has slope  $-a$ .

### Example 1

Describe the transformations and graph the function  $f(x) = -2|x - 3| + 4$ .



#### Transformations:

- Vertex: \_\_\_\_\_
- Shape: opens \_\_\_\_\_
- Branch Slopes: \_\_\_\_\_ (right), \_\_\_\_\_ (left)

### Solving Absolute Value Equations

To solve an equation containing an absolute value, first **isolate** the absolute value expression:

$$|u| = c$$

- If  $c < 0$ : There is \_\_\_\_\_ (distance cannot be negative).

- If  $c = 0$ : Solve the single case: \_\_\_\_\_.
- If  $c > 0$ : Split into two equations: \_\_\_\_\_.

**Example 2**

Solve the equation:  $2|3x - 1| + 5 = 15$

**Solving Absolute Value Inequalities**

For any algebraic expression  $u$  and a positive real number  $c$ :

- **Less-Than** (“Less-and”):

$$|u| < c \Rightarrow \text{_____}$$

This yields a single, bounded interval.

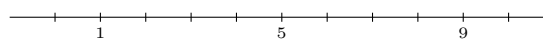
- **Greater-Than** (“Great-or”):

$$|u| > c \Rightarrow \text{_____}$$

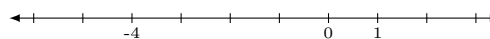
This yields a union of two unbounded intervals.

**Example 3**

Solve the inequality and graph the solution:  $|x - 5| < 4$

**Example 4**

Solve the inequality and graph the solution:  $|2x + 3| \geq 5$



*Homework:* Section 1.6 Homework

**1.7 Inverse Functions****Objectives**

- Verify inverse functions using composition.
- Find inverse functions algebraically.

- Identify one-to-one functions using the horizontal line test.
- Restrict domains to find inverse functions.

### Inverse Functions Swap Domain & Range

An **inverse function** undoes the action of the original function. If  $f$  maps  $a$  to  $b$ , then the inverse function  $f^{-1}$  maps  $b$  to  $a$ .

- **Notation:** The inverse of  $f(x)$  is written as  $f^{-1}(x)$  (read as “ $f$  inverse of  $x$ ”). Note: this is **not** an exponent, i.e.,  $f^{-1}(x) \neq \frac{1}{f(x)}$ .
- **Domain and Range:**

$$\text{Domain of } f^{-1} = \text{Range of } f \quad \text{and} \quad \text{Range of } f^{-1} = \text{Domain of } f$$

- **Coordinates:** If the point  $(a, b)$  lies on the graph of  $f(x)$ , then the point \_\_\_\_\_ lies on the graph of  $f^{-1}(x)$ .

### How to Find an Inverse Function Algebraically

To find the formula for the inverse function  $f^{-1}(x)$  of a one-to-one function  $f(x)$ :

1. Replace  $f(x)$  with  $y$ .
2. Interchange the variables  $x$  and  $y$ .
3. Solve the resulting equation for  $y$ .
4. Replace  $y$  with  $f^{-1}(x)$  (and state any domain restrictions).

#### Example 1

Find the inverse function of  $f(x) = x^3 - 4$ .

#### Example 2

Find the inverse function of  $f(x) = \frac{3x+1}{x-2}$ .

**Example 3**

Verify that  $f(x) = \frac{4}{x-3}$  and  $g(x) = \frac{4}{x} + 3$  are inverse functions of each other by showing that  $f(g(x)) = x$  and  $g(f(x)) = x$ .

**Graphical Relationship of Inverses**

The graphs of  $f(x)$  and  $f^{-1}(x)$  are reflections of each other across the identity line \_\_\_\_\_. Folding the coordinate plane along this line places the graph of the function exactly onto its inverse.

**Horizontal Line Test & One-to-One**

- A function is **one-to-one** if each output in the range corresponds to \_\_\_\_\_ in the domain.
- **Horizontal Line Test (HLT)**: A function has a function inverse if and only if its graph intersects every horizontal line in \_\_\_\_\_.

**Example 4**

Determine whether each function is one-to-one and thus has an inverse function.

a)  $f(x) = x^2 - 3$

b)  $g(x) = x^3$

c)  $h(x) = \sqrt{x}$

**Restricting Domains to Force Invertibility**

If a function is not one-to-one, it does not have an inverse function. However, we can **restrict the domain** of the original function to an interval where it is strictly increasing or decreasing (and thus one-to-one) to define an inverse function over that interval.

**Example 5**

The function  $f(x) = (x-2)^2$  is not one-to-one. Restrict its domain to  $[2, \infty)$  and find its inverse function.